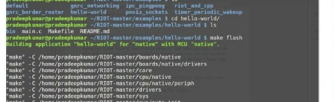

Figure 3: Running a 'hello-world' example

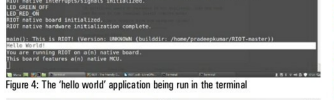

Figure 4: The 'hello world' application being run in the terminal

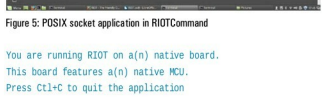

Figure 5: POSIX socket application in RIOTCommand

```
You are running RIOT on a(n) native board.
This board features a(n) native MCU.
Press Ctl+C to quit the application
```

make flash will compile and show the text and the size of the data occupied by the memory. Since this application is run under native mode, the output can be viewed using the *make term* command (refer Figures 3 and 4).

Networking applications in RIOT

There are a few applications available in RIOT. One such app is a socket application, which works with the POSIX API with UDP as the protocol. All the relevant source code is available in the directory itself.

```
pradeepkumar $ ] cd RIOT-Master/examples/posix-socket/
pradeepkumar $ ] make flash
pradeepkumar $ ] sudo make term
output of the above command
/home/pradeepkumar/RIOT-master/examples/posix_sockets/bin/
native/posix_sockets.elf tap0
RIOT native interrupts/signals initialized.
LED_GREEN_OFF
LED_RED_ON
RIOT native board initialized.
RIOT native hardware initialization complete.
main(): This is RIOT! (Version: UNKNOWN (builddir: /home/
pradeepkumar/RIOT-master))
RIOT socket example application
All up, running the shell now
> help

udp                 send data over UDP and listen on UDP
ports
reboot              Reboot the node
ps                      Prints information about running
threads.
random_init         initializes the PRNG
random_get          returns 32 bit of pseudo randomness;
ifconfig            Configure network interfaces
txtsnd              send raw data
ncache              manage neighbor cache by hand
routers             IPv6 default router list
```

Figure 5 shows the above configuration; routers can be configured, text messages can be sent and details about running processes can be searched using this application.

There are only a limited number of OSs available for IoT sensors and, among these, RIOT really has a well-documented API with huge support for most of the boards, architecture and sensors. Its main advantages are that it is actively developed and maintained, and there is an incremental version every month. Students and electronics hobbyists/enthusiasts are using Arduino and Raspberry Pi for their IoT applications these days, and RIOT really helps these boards and sensors to achieve a low memory footprint and high energy efficiency without any compromises on the performance of the IoT systems. END

References

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